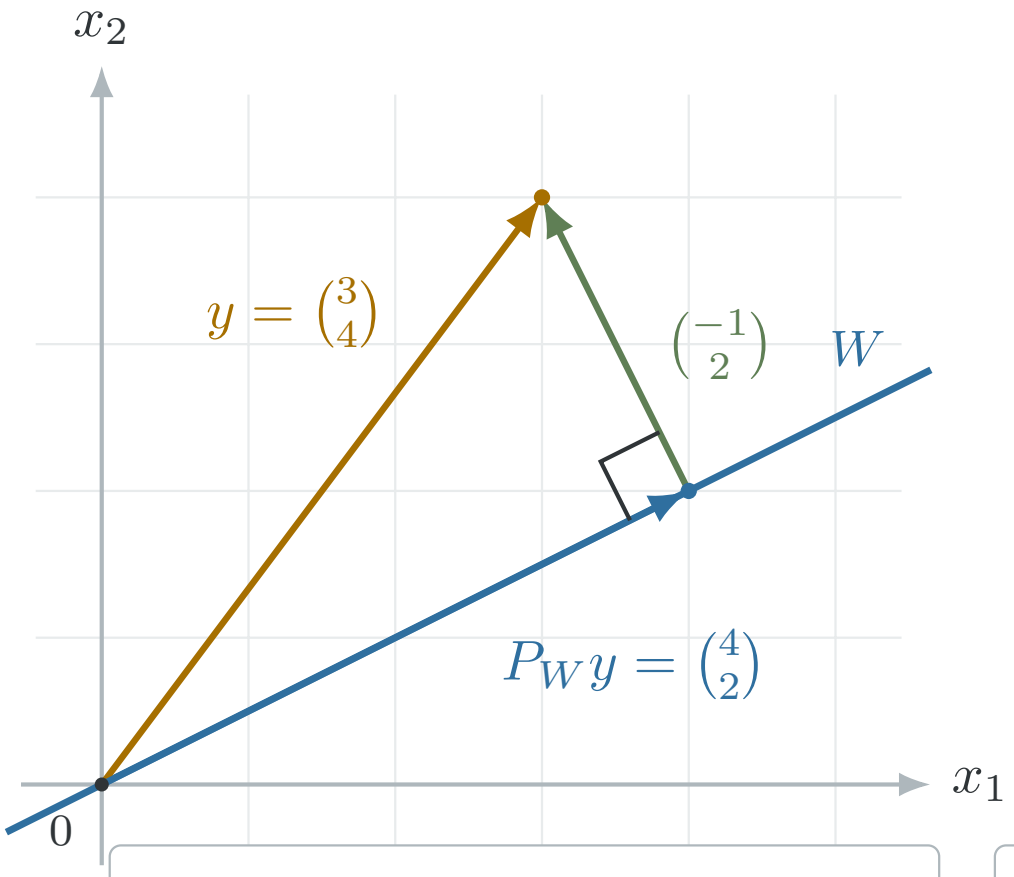
orthogonal projection onto a line in $\mathbb{R}^2$



$$q = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$P_W = qq^\mathsf{T} = \frac{1}{5} \begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix}$$

$$P_W y = \frac{1}{5} \begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}$$

$$\begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \end{pmatrix} + \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

$y =$ projection $+$ residual

$$\begin{pmatrix} 4 & 2 \end{pmatrix} \begin{pmatrix} -1 \\ 2 \end{pmatrix} = -4 + 4 = 0$$

the residual is orthogonal to W